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PEER TO PEER VALUE TRANSFER

Abstract

System and methods and computer program code are provided to transfer value from a first wallet application on a first mobile device to a second wallet application on a second mobile device. Pursuant to some embodiments, the transfer is performed over a short or medium range communications protocol and does not require for either or both devices to be “online” or connected to a remote network device or system.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a Divisional Application of and claims the benefit of U.S. patent application Ser. No. 17/453,584 filed on Nov. 4, 2021 and which claims the benefit of U.S. Provisional Patent Application No. 63/114,216 filed on Nov. 16, 2020, the contents of which provisional application are hereby incorporated by reference for all purposes.

BACKGROUND

[0002] More and more people prefer to use electronic methods of payment rather than physical cash. Unfortunately, many electronic methods of payment (such as the use of credit cards and credit card readers, or online transactions such as those provided by PayPal® or Venmo®) require that one or both of the participants in a transaction be “online” or connected to a remote network device or system. Physical cash methods of payment also have a high level of privacy, with the transaction typically known only to the parties exchanging the payment. It would be desirable to provide systems and methods for conducting transfers of value that offer the convenience and advantages of electronic payments while preserving any of the benefits associated with physical cash payments.

[0003] It would be desirable to provide systems and methods for conducting transfers of value even where there is no network connection available or where such network connection is slow or unreliable. It would further be desirable to provide such systems and methods that maintain high level of privacy and are both secure and easy to operate.

SUMMARY OF THE INVENTION

[0004] According to some embodiments, systems, methods, apparatus, computer program code and means are provided to transfer value from a first wallet application on a first mobile device to a second wallet application on a second mobile device. In some embodiments, the first mobile device receives a message from a second mobile device including information identifying a second wallet application on the second mobile device, and a wallet application on the first mobile device confirms the validity of the second wallet application. The first mobile device transmits a request to transfer an amount of value from the first wallet application to the second wallet application and sets a timer. If a confirmation is received from the second wallet application before expiration of the timer, then the transfer of value to the second wallet application is finalized.

[0005] In some embodiments, the communication session between the first and second mobile device is over a short or medium range communication protocol such as Bluetooth, NFC, or WiFi. In some embodiments, the mobile devices are in an off-line mode of operation during the transaction. In some embodiments, a status of the first wallet application is set to be active after confirming the validity of the second wallet application. In some embodiments, until both wallet applications have been set to be active, no transfer of value may take place.

[0006] In some embodiments, the information identifying the second wallet application is an encrypted hash of a wallet identifier, and the method further includes decrypting the encrypted hash to obtain a hash of the wallet identifier and comparing the hash of the wallet identifier to a list of valid hashes.

[0007] In some embodiments, finalizing the transfer of value includes incrementing a balance of the second wallet application while reducing the balance of the first wallet application by the value.

[0008] Some technical effects of some embodiments of the invention are improved and computerized ways to securely validate wallet applications with which each wallet may interact. A wallet that has not been placed into an active state is unable to send or receive value. With these and other advantages and features that will become hereinafter apparent, a more complete understanding of the nature of the invention can be obtained by referring to the following detailed description and to the drawings appended hereto.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. **1** is a high-level block diagram of a system according to some embodiments.

[0010] FIG. **2** is a flow diagram depicting a value transfer process according to some embodiments.

[0011] FIG. **3** is a flow diagram a value transfer process according to some embodiments.

[0012] FIG. **4** is a block diagram of an apparatus in accordance with some embodiments of the present invention.

DETAILED DESCRIPTION

[0013] The present invention provides significant technical improvements to interactions requiring the transfer of value between two entities operating computing devices. Pursuant to some embodiments, value may be transferred even in situations where the computing devices are "off-line" (or are otherwise unable to communicate with a central issuing authority or payment network). The present invention is directed to more than merely a computer implementation of a routine or conventional activity previously known in the industry as it significantly advances the technical efficiency, access and/or accuracy of interactions between devices by implementing a specific new method and system as defined herein. The present invention is a specific advancement in the area of peer to peer transfer and provides benefits in security, privacy and the ability to easily validate participants in the peer to peer transfer even in situations where the participants are unable to communicate with a network or other authority.

[0014] For convenience and ease of exposition, a number of terms are used herein. For example, the term "payor" is used herein to refer to an entity (such as an individual) operating a computing device to transfer value to a "payee" (an entity such as an individual operating a second computing device). The "payor" and "payee" may be individuals or they may be merchants or other entities that wish to send and receive value using features of the present invention.

[0015] As used herein, the term "digital wallet" refers to an application installed on a "mobile device" that allows a user to conduct peer to peer transactions with other digital wallets configured pursuant to the present invention. Each digital wallet may store some indication of an amount (or balance) of "value" contained within the digital wallet. The value may be, for example, a currency amount, an amount of points or tokens or the like.

[0016] As used herein, a "mobile device" may be a portable device that can be transported and be operated by a user, and may include one or more electronic components (e.g., an integrated chip, etc.). A mobile device according to some embodiments may be in any suitable form including, but not limited to a mobile phone (e.g., smart phone, cellular phone, etc.), a tablet computer, a portable media player, a personal digital assistant device (PDA), a wearable communication device (e.g., watch, bracelet, glasses, etc.), an electronic reader device, a laptop, a netbook, an ultrabook, etc. A portable communication device may also be in the form of a vehicle (e.g., a car) equipped with communication capabilities.

[0017] Prior to discussing features of the present invention in detail, an illustrative example will first be presented. The illustrative example is not intended to be limiting in any way and is simply provided for explanatory purposes. In the illustrative example, two individuals have mobile devices that have been configured with software (or "wallets") pursuant to the present invention. One of the individuals ("payor") has interacted with her bank to load her wallet with value (in this example, she has loaded $50 of value onto her wallet). The other individual ("payee") has not loaded any value onto his wallet. The payor wishes to send payee $10, but the two mobile devices are in a location or situation where neither device has cellular data access or Internet connectivity.

Embodiments of the present invention allow the two individuals to interact via local communication technologies such as via near field communication ("NFC"), Bluetooth, WiFi or the like and to transfer value as will now be described.

[0018] FIG. **1** is a high-level block diagram of a system **100** according to some embodiments of the present invention. As shown, the system **100** includes several entities that may be involved in a value transfer transaction pursuant to the present invention, including a payor device **110**, a payee device **120** as well as a network **130**, a funding service provider **150** and a wallet service provider **140**. As will be described further herein, only the payor device **110** and the payee device **120** are involved in most transactions. However, in some situations, such as when value is loaded or withdrawn from a device **110**, **120** a network **130** and one or more funding service provider devices **150** may be involved. Further, in some situations, such as when a device **110**, **120** requires an update of data or other information, a network **130** and one or more wallet service provider devices **140** may be involved in a transaction.

[0019] Prior to conducting a transaction, each of the users may enroll their respective mobile devices **110**, **120** to participate in the transaction service of the present invention by downloading a wallet application **112**, **122** onto their mobile device **110**, **120**. The information regarding the mobile device **110**, **120**, the wallet application **112**, **122** and the user may be stored at, or be accessible to, the wallet service provider **140**. Pursuant to some embodiments, each wallet application **112**, **122** has a unique identifier (referred to herein as the "wallet ID"). For example, as used herein, the wallet ID of wallet **112** is "A" and the wallet ID of wallet **122** is "B". In use, the wallet IDs will be selected to have sufficient alphanumeric characters to ensure that all wallet IDs are unique. The wallet IDs may be issued or assigned by an entity such as, for example, the wallet service provider **140** and is used to uniquely identify each wallet as will be described further herein.

[0020] Each wallet application **112**, **122** may be configured with one or more cryptographic algorithms and related data. For example, pursuant to some embodiments, each wallet application **112**, **122** is configured with a standardized hashing algorithm (that is standardized across all the wallet applications **112**, **122** and wallet service providers **140** participating in the transaction system of the present invention). The hashing algorithm is configured to be able to receive a wallet ID from another wallet (e.g., wallet application **122** may receive a wallet ID from wallet application **112**) and process it to achieve a specific unique hash. One suitable hashing algorithm that may be used is the SHA-256 algorithm (described in NIST FIPS PUB **180-2**) although other similar hashing algorithms may be used so long as the algorithm guarantees that, provided the same input the same output will be generated. As used herein, the hashed version of a wallet ID will be referred to as the hash (wallet ID) (so, the hashed version of wallet **112** is hash (A)), and the hashed versions may be generally referred to as "hashed keys" or "hashed IDs".

[0021] Each wallet application **112**, **122** is also configured with a public key cryptography system that allows each participating wallet application **112**, **122** to exchange information with other participants in the system **100** using a standard public/private key encryption mechanism. In general, each wallet application **112**, **122** may be able to encrypt its wallet ID for transmission to another participating wallet application **112**, **122** such that the recipient wallet application is able to decrypt the wallet ID as will be described further herein.

[0022] Each wallet application **112**, **122** is also configured with local storage that allows, for example, the storage of an exhaustive list of all the hashed keys of other wallet applications in the system. This list may be provided to the wallet applications **112**, **122** from one or more wallet service providers **140**, and may be initially provided when the wallet application **112**, **122** is installed on the mobile device **110**, **120**. The list may be updated periodically when a mobile device **110**, **120** is online and able to interact with a wallet service provider **140**. The list is updated to include any new wallets and to delete any cancelled or terminated wallets. Pursuant to some embodiments, only the hashed keys are included in the list to ensure that any malicious actors are unable to obtain wallet IDs by compromising local encryption. Pursuant to some embodiments, the number of wallet keys included in the list stored in each wallet application **122**, **122** may depend on the number of participating users in the system. In some embodiments, a financial institution may issue wallet applications to its customers allowing the customers to conduct peer to peer value transfers to other customers. An example of such embodiment would be issuing wallet applications to a set of customers expected to temporarily reside in a location with no or highly latent network connection, e.g. traveling in space. In some embodiments, an entire region or country may allow any user to conduct peer to peer value transfers to other users in the region or country (in which case, each wallet application will store a list of all users in that region or country). Embodiments allow a large number of users to conduct peer to peer transactions with each other, and the list need not be limited by financial institution, region, or country.

[0023] As shown in FIG. **1**, the two mobile devices **110**, **120** are interacting via a series of messages (shown as messages **1-4**) to conduct a transfer of value. Continuing the illustrative example introduced above, the mobile device **110** is operated by a payor who wishes to send $10 to a payee who is operating mobile device **120**. Both devices have been configured to have installed wallet applications **112**, **122** allowing them to participate in a value transfer of the present invention. While the details of the messages and processing performed by each of the mobile devices **110**, **120** will be described in further detail below in conjunction with FIGS. **2-3**, a brief overview will now be provided.

[0024] In the transaction shown in FIG. **1**, the payor initiates the transfer by sending information to the payee in message “1”. The payee uses information from the first message to confirm the validity of the wallet **112**. By first confirming the validity of the wallet **112**, the payee can avoid interacting with any mobile devices and wallet applications that are fraudulent or unauthorized. If the validity is confirmed, the payee wallet changes its state to “active”, allowing the transaction to proceed, and the wallet application **122** causes a message “2” to be sent to the payor. Message “2” includes information identifying wallet application **122** which is used by wallet application **112** to confirm the validity of the payee wallet application **122**. If the validity is confirmed, the payor wallet application **112** causes a further message “3” to be sent which includes information about the amount of value to be transferred as well as a random code such as a short 3-5 digit PIN. The payee wallet application **122** processes the received message and sends a confirmation message “4” which causes the payor wallet application **112** to reduce its balance by the amount of the value and causes the payee wallet application **122** to be increased by the amount of the received value. The payee may then use that received value in subsequent transactions to transfer some or all of the value to other participants in the system of the present invention. The payee may also convert the value to a fiat currency (such as US dollars) by interacting with an entity such as a funding service provider **150** to withdraw the value from the wallet.

[0025] Pursuant to some embodiments, the funding service provider **150** and the wallet service provider **140** may be a payment network service provider such as, for example, Mastercard International Incorporated (referred to herein as “Mastercard”). The funding service provider **150** may be a financial institution that is able to interact with wallet applications **112**, **122** of the present invention (e.g., by confirming the validity of each wallet by validating hashes and encrypted hashes as described herein) and is able to conduct peer to peer value transfers with those wallet applications. Interactions with a funding service provider **150** may result in value being exchanged into (or from) currency or other tangible forms of value. For example, the $10 of value on the payor's wallet application **112** at the start of the transaction may have been transferred onto that wallet application **112** from the payor's bank account at funding service provider **150**. Funds may be transferred from credit cards, bank accounts, or other sources of value such as PayPal or the like.

[0026] Mobile devices according some embodiments can be configured to communicate with external entities (such as wallet service providers **140**, funding service providers **150** or the like) through long range communications technologies and protocols such as cellular communication and Internet protocols (represented by network **130** in FIG. **1**). They may also be configured to communicate with other devices (such as other mobile devices **110**, **120**) using any suitable short or medium range communications technology including Bluetooth (classic and BLE-Bluetooth low energy), NFC (near field communications), IR (infrared), Wi-Fi, etc. In general, as used herein, when longer range communications are available the mobile device will be referred to as “on-line” and such communications are not available the mobile device will be referred to as “off-line” (but still able to communicate with other mobile devices using short or medium range technologies).

[0027] Some or all of the processing described herein may be performed automatically or otherwise be automated by one or more computing devices or systems. As used herein, the term “automate” may refer to, for example, actions that can be performed with little (or no) intervention by a human.

[0028] Further details of peer to peer value transfer transactions pursuant to some embodiments will now be described by reference to FIG. **2** (where a value transfer process **200** is shown from the perspective of the payor mobile device **110**) and FIG. **3** (where a value transfer process **300** is shown from the perspective of the payee mobile device **120**). Similar processes may be used to load or unload value into wallet applications in transactions involving funding service providers **150** and will not be described separately herein.

[0029] Referring first to FIG. **2**, a value transfer transaction process **200** begins at **202** where the wallet application **112** of the payor mobile device **110** is operated to transmit a request message to the payee wallet application **122** of the payee mobile device **120**. Although not shown, the payee and payor mobile devices **110**, **120** may already be in communication using Bluetooth or some other form of communication. Alternatively, in some embodiments, the devices may come into communication as part of process **200** (e.g., at **202**). In some embodiments, the request message at **202** may include information identifying the payor wallet application **112**, such as the payor wallet ID (which is subsequently used by the payee wallet application **122** to validate the authenticity of the payor wallet application **112** as will be described further below in conjunction with FIG. **3**). In some embodiments, the request message includes an encrypted hash associated with the payor wallet application **112** (referred to as hash (A)). The payee wallet application **122** performs processing in response to the request (as will be described further in conjunction with FIG. **3**). If the processing at the payee device is successful, processing at the payor mobile device **110** continues at **204**.

[0030] Processing continues at **204** where the payor wallet application **112** of the payor mobile device **110** receives a response from the payee wallet application **122** including information identifying the payee wallet application **122** (such as the payee wallet ID). In some embodiments, the response message at **204** may include an encrypted hash associated with the payee wallet application **122** (referred to as hash (B)). Processing continues at **206** where the payor wallet application **112** operates to use a hashing algorithm (the algorithm that is used by all of the devices in the system **100** as described above) to generate a hash using the payee wallet ID. Because the same hashing algorithm is used by both the payor wallet application **122** and the wallet service provider **140**, the hash generated by the payor wallet application **122** and the wallet service provider **140** should be identical.

[0031] A determination is then made at **208**. If the hash generated by the payor wallet application **112** was not found in the list of hashes, processing continues at **210** where the wallet application **112** cancels the value transfer process (as the payee wallet application **122** is not an application that the wallet application **112** is permitted to transact with). If, however, the hash was found in the list of hashes, processing continues at **212** where the wallet application **112** operates to change the wallet state to “active”. Pursuant to some embodiments, an “active” state means that the wallet is ready to send or receive funds from another wallet. If either wallet in a transaction is in an “inactive” state, no value transfer can occur. Processing at **208** also includes placing a “hold” on the amount of value to be transferred to the payee. For example, in the illustrative example, the $10 to be transferred from wallet A will be marked as “unavailable” or “blocked”. In this way, unless the transaction is aborted or canceled as will be described further herein, the funds are no longer available in wallet A.

[0032] In some embodiments, where the response received at **204** included an encrypted hash of the payee wallet ID, processing at **206** and **208** may include operating the payor wallet application **112** to decrypt the encrypted hash (B) and attempts to locate the hash (B) in the list of hashes stored in local storage (where the presence of a hash (ID) on the list in local storage indicates that the wallet application associated with the hash (ID) is an authorized and valid wallet application). In either event, (whether an encrypted hash or a clear text wallet ID is received at **204**), embodiments provide a secure approach to validating the authenticity of a counterparty wallet application in each transaction.

[0033] The payee wallet application **122** processes the message from the payor wallet and is expected to return a confirmation message. Because the confirmation message may not arrive immediately, a timer is started when the request at **214** is transmitted. If the payee does not respond with a confirmation before the timer expires, the value transfer is canceled. For example, if the timer expires before confirmation at **216**, processing continues at **218** where the payor wallet application **112** removes or cancels the “hold” on the value. In the example, the $10 in wallet A is marked as “available” again for the payor's use. Processing continues at **210** and the value transfer transaction is canceled.

[0034] If, however, a confirmation message is received before the timer expires at **216**, processing continues at **220** where the payor wallet application **112** causes the random PIN to be displayed on a display device of mobile device **110**. The display may also prompt the user of mobile device **110** (the payor) to communicate the random PIN to the payee (the operator of mobile device **120**). This ensures the finality of the funds transfer. As will be discussed further below, the payee enters the PIN into their mobile device **120** to finalize the transaction. Pursuant to some embodiments, the random PIN is not displayed or known to either user until 220 to reduce the risk of fraud. Until this point, the random PIN was only known to the wallet applications **112**, **122**. While a random PIN is referred to herein, those skilled in the art, upon reading the present disclosure, will appreciate that the random PIN may be any semi-random code generated that is preferably difficult to generate or anticipate and that is relatively short in length (e.g., such as a code having four or five digits or characters).

[0035] From the perspective of the payor mobile device **110** and the payor wallet application **112**, the value transfer transaction **200** is completed at **222** and the wallet application **112** balance is permanently reduced by the amount of value transferred.

[0036] Reference is now made to FIG. **3** where a value transfer transaction **300** from the perspective of a payee mobile device **120** and payee wallet application **122** is shown. Again, as will the process **200**, processing assumes that two devices (such as two mobile devices **110**, **120** or a mobile device and a funding service provider **150**) are interacting and that both devices have a wallet application **112**, **122** (or an equivalent in the case of a funding service provider **150**) stored thereon. Process **300** begins at **302** where the payee wallet application **122** receives a request from a payor wallet application **112** which includes information identifying the payor wallet application **112** such as a payor wallet ID. In some embodiments, the information identifying the payor wallet application **112** may be, for example, an encrypted hash of the payor's wallet ID (hash (A)). At **304**, the payee wallet application **122** performs processing to retrieve or generate a hash of the payor wallet ID by either using a hashing algorithm in conjunction with the payor wallet ID to generate the hash or by decrypting the encrypted hash (e.g., using a private key that corresponds to the public key used to encrypt the hash) to produce an unencrypted hash of the payor wallet ID. Processing continues at **306** where the payee wallet application **122** attempts to match the generated or unencrypted hash of the payor wallet ID with a hash in the local storage list of hashes stored in the payee device **120**.

[0037] If the search for a matching hash is unsuccessful, processing continues at **308** where the value transfer transaction **300** is canceled. That is, the transaction is not able to be completed because the payee wallet application **122** was unable to confirm that the payor wallet application **112** was a legitimate wallet application. If, however, the processing at **306** results in the unencrypted hash being found within the list of hashes (confirming the validity of the payor wallet application **112**), processing continues at **310** where the payee wallet application **122** changes the wallet state from “inactive” to “active” and transmits a response to the payor wallet application **112**. The response to the payor wallet application **112** may include an encrypted wallet hash of the payor wallet (e.g., hash (B)). The response also includes information indicating that the payee wallet is now in an “active” state and ready to receive a value transfer.

[0038] The payor wallet application **112** performs processing (as described in FIG. **2** at **203-212**) and if the payor was able to confirm the validity of the payee wallet application **122**, processing continues at **312** where the payee wallet application **122** receives a request from the payor wallet application **112** with an indication of the amount of value to be transferred as well as a randomly generated PIN. If the payee wallet application **122** receives the request (and is not turned off or otherwise unable to receive the request or send a confirmation message), the payee wallet application **122** sends a confirmation message back to the payor wallet application **112** at **314**. The confirmation message confirms that the payee has received the request to receive funds and that the mobile device **120** is not turned off or unable to receive the request (or send the response). If the confirmation message is not transmitted promptly (e.g., within five seconds), the transfer may be canceled.

[0039] If the confirmation message was sent at **314**, processing continues at **316** where the payee wallet application **122** prompts the user (the payee) to enter the randomly generated PIN provided by the user of the payor device **110** (e.g., at step **220** of FIG. **2**). In some embodiments, the PIN must be entered within a set amount of time (e.g., 1 minute) to reduce risk of fraud. If the mobile device **120** turns off unexpectedly (e.g., due to a low battery), the timer may be continue from the time the device turned off, allowing the user time to enter the PIN. Processing continues at **318** where a determination is made whether the entered PIN is the same as the PIN received at **312**. If not, processing continues at **308** and the value transfer is canceled. If the entered PIN is the same as the received PIN, processing continues at **320** where the balance of the payee wallet is increased by the amount of the value and the value transfer process is completed.

[0040] The embodiments described herein may be implemented using any number of different hardware configurations. FIG. **4** illustrates a mobile device **400** that may be used in any of the methods and processes described herein, in accordance with an example embodiment. For example, the mobile device **400** may be operated as either the payor device **110** or the payee device **120** of FIG. **1**.

[0041] In some embodiments, device **400** can include some or all of the components described with respect to FIGS. **1-3**. Device **400** has a bus **412** or other electrical components that operatively couple an input/output (“I/O”) section **414** with one or more computer processors **416** and memory section **418**. I/O section **414** can be connected to a display **404**, which can have one or more aspects such as a touch-sensitive component (not shown). In addition, I/O section **414** can be connected with communication unit **430** for receiving application and operating system data, using Wi-Fi, Bluetooth, near field communication (NFC), cellular, and/or other wireless communication techniques (e.g., to allow the device **400** to interact with other mobile devices to conduct transactions as described herein or to interact with funding service providers **150** and wallet service providers **140**). Mobile device **400** can include one or more input mechanisms **406** such as a keypad, a button, a touch-screen display, or the like.

[0042] Input mechanism **408** is, optionally, a microphone, in some examples. Mobile device **400** optionally includes various sensors (not shown), such as a GPS sensor, accelerometer, directional sensor (e.g., compass), gyroscope, motion sensor, and/or a combination thereof, all of which can be operatively connected to I/O section **414**.

[0043] Memory section **418** mobile device **400** can include one or more non-transitory computer-readable storage mediums, for storing computer-executable instructions, which, when executed by one or more computer processors **416**, for example, can cause the computer processors to perform the techniques described below, including processes **200** and **300**. A computer-readable storage medium can be any medium that can tangibly contain or store computer-executable instructions for use by or in connection with the instruction execution system, apparatus, or device. In some examples, the storage medium is a transitory computer-readable storage medium. In some examples, the storage medium is a non-transitory computer-readable storage medium. The non-transitory computer-readable storage medium can include, but is not limited to, magnetic, optical, and/or semiconductor storages. Examples of such storage include magnetic disks, optical discs based on CD, DVD, or Blu-ray technologies, as well as persistent solid-state memory such as flash, solid-state drives, and the like. Mobile device **400** is not limited to the components and configuration of FIG. **4** but can include other or additional components in multiple configurations.
[0044] Memory section **418** may store one or more applications **422***a-n* including, for example, a wallet application **112**, **122** as described herein. Memory section **418** may also provide local storage for the storage of a list of wallet keys as described herein, as well as storage for one or more cryptographic applications.
[0045] As will be appreciated based on the foregoing specification, the above-described examples of the disclosure may be implemented using computer programming or engineering techniques including computer software, firmware, hardware or any combination or subset thereof. Any such resulting program, having computer-readable code, may be embodied or provided within one or more non-transitory computer-readable media, thereby making a computer program product, i.e., an article of manufacture, according to the discussed examples of the disclosure. For example, the non-transitory computer-readable media may be, but is not limited to, a fixed drive, diskette, optical disk, magnetic tape, flash memory, external drive, semiconductor memory such as read-only memory (ROM), random-access memory (RAM), and/or any other non-transitory transmitting and/or receiving medium such as the Internet, cloud storage, the Internet of Things (IoT), or other communication network or link. The article of manufacture containing the computer code may be made and/or used by executing the code directly from one medium, by copying the code from one medium to another medium, or by transmitting the code over a network.
[0046] The computer programs (also referred to as programs, software, software applications, “apps”, or code) may include machine instructions for a programmable processor and may be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, apparatus, cloud storage, internet of things, and/or device (e.g., magnetic discs, optical disks, memory, programmable logic devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a machine-readable medium that receives machine instructions as a machine-readable signal. The “machine-readable medium” and “computer-readable medium,” however, do not include transitory signals. The term “machine-readable signal” refers to any signal that may be used to provide machine instructions and/or any other kind of data to a programmable processor.
[0047] The above descriptions and illustrations of processes herein should not be considered to imply a fixed order for performing the process steps. Rather, the process steps may be performed in any order that is practicable, including simultaneous performance of at least some steps. Although the disclosure has been described in connection with specific examples, it should be understood that various changes, substitutions, and alterations apparent to those skilled in the art can be made to the disclosed embodiments without departing from the spirit and scope of the disclosure as set forth in the appended claims.

Claims

1. A computerized method to operate a first mobile device having a first wallet application to act as a payee in a transaction, the method performed by the first mobile device, comprising: receiving a message from a second mobile device including information identifying a second wallet application on the second mobile device; confirming the validity of the second wallet application; transmitting, to the second mobile device, an indication that a status of the first wallet application has been set to be active; receiving a message from the second mobile device having a code and an amount of value to be transferred; confirming that a code entered by a user into the first mobile device matches the code received from the second mobile device; and increasing a balance of the first wallet application by the amount of value.
2. The method of claim 1, confirming the validity of the second wallet application further comprises: determining that the information identifying the second wallet application matches information in a locally-stored list of valid wallet applications.
3. The method of claim 1, further comprising: establishing a communication session with the second mobile device over a short or medium range communication protocol.
4. The method of claim 3, wherein the communication protocol is at least one of: Bluetooth, NFC, and WiFi.
5. The method of claim 1, wherein receiving a message from the second mobile device further includes receiving a received hash of the information identifying the second wallet application.
6. The method of claim 5, wherein confirming the validity of the second wallet application further comprises: generating, by the first mobile device using the first wallet application, a hash of the information identifying the second wallet application; and confirming a validity of the second wallet application by confirming that (i) the generated hash matches the received hash, and (ii) the generated hash is in a list of hashes stored on the mobile device, wherein the list of hashes includes the unique hash for each of the plurality of wallet applications in a payment system and wherein the list of hashes is provided to the first mobile device from a wallet service provider.
7. The method of claim 1, wherein confirming that a code entered by a user into the first mobile device matches the code received from the second mobile device further comprises: confirming that the code is entered by the user into the first mobile device within a timer period.
8. The method of claim 1, further comprising: finalizing the transaction by causing a balance of the second wallet application to be reduced by the amount of value.
9. A non-transitory computer-readable storage medium comprising executable instructions for use in operating a first mobile device having a first wallet application to act as a payee in a transaction, which, when executed by at least one processor of the first mobile device, cause the at least one processor to: receive a message from a second mobile device including information identifying a second wallet application on the second mobile device; confirm the validity of the second wallet application; transmit, to the second mobile device, an indication that a status of the first wallet application has been set to be active; receive a message from the second mobile device having a code and an amount of value to be transferred; confirm that a code entered by a user into the first mobile device matches the code received from the second mobile device; and increase a balance of the first wallet application by the amount of value.
10. The non-transitory computer-readable storage medium of claim 9, wherein confirming the validity of the second wallet application further comprises executable instructions, when executed by the at least one processor, further cause the at least one processor to: determine that the information identifying the second wallet application matches information in a locally-stored list of valid wallet applications.
11. The non-transitory computer-readable storage medium claim 9, further comprises executable instructions, when executed by the at least one processor, further cause the at least one processor

to: establish a communication session with the second mobile device over a short or medium range communication protocol.

12. The non-transitory computer-readable storage medium claim 11, wherein the communication protocol is at least one of: Bluetooth, NFC, and WiFi.

13. The non-transitory computer-readable storage medium of claim 9, wherein receiving a message from the second mobile device further includes receiving a received hash of the information identifying the second wallet application.

14. The non-transitory computer-readable storage medium of claim 13, wherein confirming the validity of the second wallet application further comprises executable instructions, when executed by the at least one processor, further cause the at least one processor to: generate, by the first mobile device using the first wallet application, a hash of the information identifying the second wallet application; and confirm a validity of the second wallet application by confirming that (i) the generated hash matches the received hash, and (ii) the generated hash is in a list of hashes stored on the mobile device, wherein the list of hashes includes the unique hash for each of the plurality of wallet applications in a payment system and wherein the list of hashes is provided to the first mobile device from a wallet service provider.

15. The non-transitory computer-readable storage medium of claim 9, wherein confirming that a code entered by a user into the first mobile device matches the code received from the second mobile device further comprises: confirm that the code is entered by the user into the first mobile device within a timer period.

16. The non-transitory computer-readable storage medium of claim 9, further comprising executable instructions, when executed by the at least one processor, further cause the at least one processor to: finalize the transaction by causing a balance of the second wallet application to be reduced by the amount of value.

17. A mobile device, comprising: a communication interface configured to communicate with a second mobile device via a short or medium range communication protocol; a non-transitory computer-readable storage medium storing a mobile wallet application comprising executable instructions; and a processor, when executing the mobile wallet application, configured to: receive a message from a second mobile device including information identifying a second wallet application on the second mobile device; confirm the validity of the second wallet application; transmit, to the second mobile device, an indication that a status of the first wallet application has been set to be active; receive a message from the second mobile device having a code and an amount of value to be transferred; confirm that a code entered by a user into the first mobile device matches the code received from the second mobile device; and increase a balance of the first wallet application by the amount of value.